PROJECT NAME:

Mental Health Resource Allocator for Schools

1. Overall Approach:-

To address the challenges of unstructured mental health identification in schools, a Centralized Digital Monitoring System has been adopted. This solution replaces informal observation methods with a structured and data-driven workflow.

The core strategy is Standardized Mental Health Input Collection. By requiring structured inputs such as stress level, mood, sleep quality, and behavioral indicators, the system reduces ambiguity and enables accurate analysis.

The application features distinct interfaces:

● Student Portal:

Focuses on simple self-assessment, allowing students to input their mental health status regularly and track their well-being.

● Admin/Counselor Dashboard:

Focuses on risk analysis, student prioritization, and efficient allocation of counseling resources.

2.System Architecture Overview

The system follows a modular 3-Tier Architecture:

● **Frontend (Presentation Layer):**

○ **Student View**:Provides an easy-to-use interface for entering mental health data such as stress level, mood, sleep, and behavior. It also allows students to view their status.

○ **Admin View:**Displays a dashboard with student risk levels, categorized into Safe, Warning, and Critical. Includes tables and visual indicators for prioritization.

● **Backend Logic (Business Layer):**

○ Acts as the central controller to process and validate incoming data.

○ Implements the Risk Scoring Algorithm to analyze student inputs and determine mental health status.

○ **Manages the workflow:**

Safe → Needs Attention → Critical

● **Data Persistence Layer:**

○ Stores structured student data, risk scores, and historical records.

○ Enables tracking of behavioral trends and supports data-driven decision-making.

3. Key Components:

The system is composed of three primary logical engines:

● **Mental Health Analysis Engine:**

Converts student inputs into numerical values

Calculates risk score using weighted parameters

Classifies students into risk categories

● **Smart Risk Prioritization System:**

○ Logic:

Students are ranked based on their risk score.

○ Action:

High-risk students are prioritized for immediate counseling, while moderate-risk students are scheduled accordingly.

● **Resource Allocation Engine:**

Assigns counselors based on priority

Ensures efficient use of limited mental health resources

● **Reporting & Visualization Service:**

Displays key insights such as:

Number of high-risk students

Overall mental health trends

Uses color-coded indicators (Red, Yellow, Green)

4.Data Flow Explanation:

1. **Submission:**A student logs in and submits mental health data (stress, mood, sleep, etc.).

2.Processing: The system processes inputs and calculates a risk score using predefined weights.

3. Classification:

The student is categorized into:

Safe

Needs Attention

Critical

4.Visualization:

The dashboard updates instantly, highlighting critical students in red for urgent action.

5.Resource Allocation:

Counselors are assigned based on priority levels, ensuring timely support.

6.Continuous Monitoring:

The system tracks data over time to detect behavioral changes and worsening conditions.

5. Design Decisions :

- **Web-Based Application:**

Chosen to ensure accessibility across devices (mobile/laptop) without installation.

- **Structured Input System:**

Mandatory fields like stress level, mood, and behavior ensure consistent data collection.

- **Risk-Based Classification:**

Implemented to prioritize students effectively and avoid resource overload.

- Color-Coded Interface:

🔴 Red → Critical

🟡 Yellow → Needs Attention

🟢 Green → Safe

👉 Enables quick identification of high-risk students.